

Part A – Setup

Q1. Import the required libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
```

Part B – Create the Dataset

Q2. Create a simple dataset for salary prediction.

```
data = {
    "Years_of_Experience": [1,2,3,4,5,6,7,8,9,10],
    "Salary": [35000,37000,42000,50000,55000,61000,67000,72000,78000,83000]
}

df = pd.DataFrame(data)
print(df)
```

Part C – Split the Data

Q3. Define input (X) and output (Y), and split the dataset into training and testing sets.

Part D – Model 1: Underfitting

Q4. Train a Decision Tree model with depth = 1.

Part E – Model 2: Overfitting

Q5. Train a Decision Tree model with depth = 10.

Part F – Compare Model Complexity

Q6. Compare training and testing errors for different depths (1–10).

Q7. At what tree depth does the model perform best?

More Questions

Q8. What is underfitting?

Q9. What is overfitting?

Q10. How can we control overfitting?

Q11. Compare Training Accuracy and Testing Accuracy.

Q12. Visualize Model Predictions at Different Depths. [1,3,5 and 10]

Q13. Build a Ridge Regression model using the same dataset and compare R^2 score with Decision Tree.

Q14. How can we reduce overfitting besides reducing depth?

Q15. Why is Overfitting Dangerous in Real-World Applications?